AN2DL - Second Challenge Report
Team Name

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1 Introduction

Regarding this four-class image classification problem, we employs a patch-based approach where images are subdivided into smaller regions. These patches are processed using **Multiple Instance Learning (MIL)** aggregation techniques to produce a slide-level prediction.

2 Problem Analysis

The dataset consists of pathology images categorized into four subtypes: **Luminal A, Luminal B, HER2(+), and Triple negative**, includes 690 training images and their mask files.

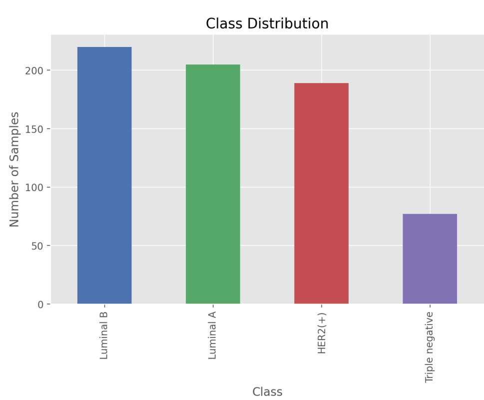

Figure 1: The distribution of classes in the dataset

The dataset exhibits class imbalance, with the **Luminal B** subtype being the most prevalent, **Triple negative** subtypes are underrepresented.

We are trying to extract multiple patches from each high-resolution pathology image.

3 Method

This section should detail your approach. You can use equations to explain your methodology. For example, a simple model representation:

$$f(x) = \mathrm{softmax}(Wx + b) \quad (1)$$

Or a more complex loss function:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} y_i \log(\hat{y}_i) \quad (2)$$

Reference these equations in your text, like:"As shown in equation 1..."

4 Experiments

For your experiments, you might want to present your results in tables. Here's an example of a wide table comparing different models:
For more specific measurements, you might use a narrower table:

Table 1: An example of wide table. Best results are highlighted in **bold**.

Model	Accuracy	Precision	Recall	ROC AUC
VGG18	72.20 ± 3.06	94.95 ± 0.52	86.95 ± 0.55	80.16 ± 0.81
Custom Model	27.71 ± 3.19	75.70 ± 1.07	55.75 ± 2.16	36.60 ± 1.26
ResNet18	**89.24 ± 2.38**	**95.54 ± 0.49**	**93.43 ± 1.30**	**91.68 ± 0.71**

Table 2: An example of table. Best results may be highlighted in **bold**.

Time [μs]	Distance [mm]
22±4	8±1
17±3	7±1
15±3	6±1
13±2	5±1
10±2	4±1
8±2	3±1
5±1	2±1
37±1	1±1

You can also include figures to visualise your results:

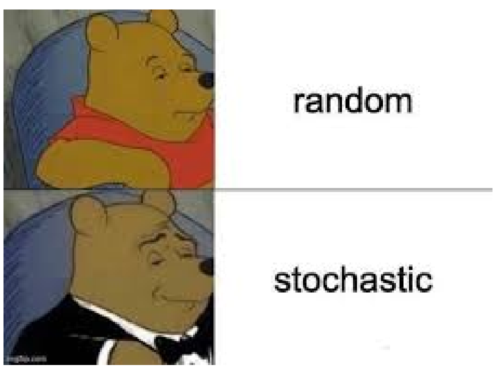

Figure 2: Example figure showing [describe what the figure shows]

Reference figures using like:"As shown in Figure 2..."

5 Results

Present your main findings here. You might want to:

- Compare your results with baselines
- Highlight key achievements using **bold text**
- Explain any unexpected outcomes

6 Discussion

In this section, analyse your results critically. Consider:

- Strengths and weaknesses
- Limitations and assumptions

7 Conclusions

Summarise your work and discuss potential future directions. This is where you can:

- Restate main contributions
- Suggest improvements
- Propose future work